

1)

The screenshot shows the IntelliJ IDEA interface with a project named 'LABA'. The left sidebar displays the project structure, including directories like 'gradle', 'idea', 'kotlin', 'build', 'src', 'main', 'test', and files like 'gitignore', 'build.gradle.kts', 'gradle.properties', 'gradlew', 'gradlew.bat', and 'settings.gradle.kts'. The main editor window shows a Kotlin file named 'phi.kt' with the following code:

```
1 fun sumList(numbers: List<Int>): Int {  
2     return numbers.sum()  
3 }  
4  
5 main() {  
6     val numbers = listOf(1, 2, 3, 4, 5, 6, 7)  
7     println("Сумма элементов списка: ${sumList(numbers)}")  
8 }
```

The Run console at the bottom shows the command: `C:\Users\atypo\jdk\openjdk-23.0.1\bin\java.exe -javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56316 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8` and the output: `Сумма элементов списка: 28`. The process finished with exit code 0.

2)

The screenshot shows the IntelliJ IDEA interface with the same project 'LABA'. The main editor window shows a Kotlin file named 'phi.kt' with the following code:

```
1 fun differenceBetweenMaxAndMin(numbers: List<Int>): Int {  
2     if (numbers.isEmpty()) return 0  
3     return numbers.maxOrNull()!! - numbers.minOrNull()!!  
4 }  
5  
6 main() {  
7     val numbers = listOf(10, 5, 20, 3, 8)  
8     println("Разность max и min: ${differenceBetweenMaxAndMin(numbers)}")  
9  
10    val emptyList = listOf<Int>()  
11    println("Разность для пустого списка: ${differenceBetweenMaxAndMin(emptyList)}")  
12 }
```

The Run console at the bottom shows the command: `C:\Users\atypo\jdk\openjdk-23.0.1\bin\java.exe -javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56361 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8` and the output: `Разность max и min: 17` and `Разность для пустого списка: 0`. The process finished with exit code 0.

3)

```

1 fun combineLists(list1: List<Int>, list2: List<Int>): List<Int> {
2     return list1 + list2
3 }
4 fun combineListsAlternative(list1: List<Int>, list2: List<Int>): List<Int> {
5     return mutableListof<Int>().apply {
6         addAll(list1)
7         addAll(list2)
8     }
9 }
10
11 fun main() {
12     val listA = listOf(1, 2, 3)
13     val listB = listOf(4, 5, 6)
14     val combined = combineLists(listA, listB)
15     println("Объединенный список: $combined")
16 }

```

Run

```

C:\Users\atypo\jdk\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56392" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Объединенный список: [1, 2, 3, 4, 5, 6]
Process finished with exit code 0

```

4)

```

1 fun isProfitable(prob: Double, prize: Double, pay: Double): Boolean {
2     return prob * prize > pay
3 }
4
5 fun isProfitableVerbose(prob: Double, prize: Double, pay: Double): Boolean {
6     if (prob * prize > pay) {
7         return true
8     } else {
9         return false
10    }
11 }
12
13 fun main() {
14     println(isProfitable(prob=0.5, prize=10.0, pay=4.0))
15     println(isProfitable(prob=0.1, prize=100.0, pay=10.0))
16     println(isProfitable(prob=2.0, prize=5.0, pay=9.0))
17 }

```

Run

```

C:\Users\atypo\jdk\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56423" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
true
false
true
Process finished with exit code 0

```

5)

```

1 fun checkProfit(prob: Double, prize: Double, pay: Double): Boolean {
2     return prob * prize > pay
3 }
4
5 fun main() {
6     println(checkProfit(0.2, 50.0, 9.0))
7     println(checkProfit(0.1, 30.0, 5.0))
8     println(checkProfit(0.5, 10.0, 15.0))
9 }

```

```

C:\Users\atypo\.jds\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56455" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
true
false
false
Process finished with exit code 0

```

6)

```

1 fun isSumLessThan100(a: Int, b: Int): Boolean {
2     return a + b < 100
3 }
4
5 fun isSumLessThan100Verbose(a: Int, b: Int): Boolean {
6     if (a + b < 100) {
7         return true
8     } else {
9         return false
10    }
11 }
12
13 fun main() {
14     println(isSumLessThan100(50, 40))
15     println(isSumLessThan100(50, 60))
16     println(isSumLessThan100(99, 0))
17     println(isSumLessThan100(100, -1))
18 }

```

```

C:\Users\atypo\.jds\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56476" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
true
false
true
true
Process finished with exit code 0

```

7)

```

1 fun isDivisibleBy100(number: Int): Boolean {
2     return number % 100 == 0
3 }
4
5 fun isDivisibleBy100(verbose: Int): Boolean {
6     if (number % 100 == 0) {
7         return true
8     }
9     return false
10 }
11
12 fun main() {
13     println(isDivisibleBy100(200)) // true
14     println(isDivisibleBy100(150)) // false
15     println(isDivisibleBy100(-300)) // true
16     println(isDivisibleBy100(0)) // true
17     println(isDivisibleBy100(99)) // false
18 }

```

Run output:

```

true
false
true
false
Process finished with exit code 0

```

8)

```

1 fun checkPowerEquality(n: Int, k: Int): Boolean {
2     return when {
3         k == 0 -> n == 1 // Короче n^0 = 1
4         k == 1 -> n == 1 // 1 в любой степени = 1
5         k == -1 -> n == -1 || n == 1 // (-1)^n = -1 или 1
6         k > 0 -> {
7             var result = 1
8             repeat(k) { result *= n }
9             result == n
10        }
11        else -> false
12    }
13 }
14
15 fun main() {
16     println(checkPowerEquality(0, 1))
17     println(checkPowerEquality(0, 2))
18     println(checkPowerEquality(0, 3))
19     println(checkPowerEquality(0, 4))
20     println(checkPowerEquality(-1, -3))
21     println(checkPowerEquality(0, 0))
22     println(checkPowerEquality(0, 1))
23     println(checkPowerEquality(0, 100))
24     println(checkPowerEquality(0, -8))
25 }

```

Run output:

```

true
true
true
true
true
true
false
false
Process finished with exit code 0

```

The screenshot shows the IntelliJ IDEA IDE interface. The top bar includes the menu bar (File, Edit, View, Window, Help) and the toolbar with icons for running, debugging, and other IDE functions. The left sidebar displays the project structure for a project named "LABA". The main editor shows a Kotlin file named "phi.kt" with the following code:

```
1 fun isKEqualToN(n: Int, k: Int): Boolean {
2     return when {
3         k == 0 -> n == 1
4         k == 1 -> n == 1
5         k == -1 -> n == -1 || n == 1
6         k > 0 -> {
7             var result = 1L
8             repeat(k) {
9                 if (result > Int.MAX_VALUE) return false
10            }
11            result == n.toLong()
12        }
13     } else -> false
14 }
15
16
17
18
19 fun main() {
20     println(isKEqualToN(1, 1))
21     println(isKEqualToN(256, 4))
22     println(isKEqualToN(3125, 5))
23     println(isKEqualToN(-1, -1))
24     println(isKEqualToN(1, -1))
25     println(isKEqualToN(1, 0))
26     println(isKEqualToN(100, 5))
27     println(isKEqualToN(Int.MAX_VALUE, 2))
28 }
```

The Run tab at the bottom shows the execution output, indicating the process finished with exit code 0.

The screenshot shows the IntelliJ IDEA IDE interface. The top bar includes tabs for 'LABA' and 'Version control'. The left sidebar shows the project structure for 'LABA' at 'C:\Users\atypol\IdeaProjects\LABA'. The main editor displays a Kotlin file 'phi.kt' with the following code:

```
1 fun repetition(txt: String, n: Int): String {
2     return when {
3         n <= 0 -> ""
4         n == 1 -> txt
5         else -> txt + repetition(txt, n - 1)
6     }
7 }
8
9
10 fun main() {
11     println(repetition("aa", 7))
12     println(repetition("kiskiki", 4))
13     println(repetition("cherry", 2))
14     println(repetition("a", 5))
15     println(repetition("test", 0))
16 }
```

The bottom panel shows the 'Run' configuration and the execution output. The output displays the results of the recursive function:

```
C:\Users\atypol\jdk-23.0.1\bin\java.exe -javaagent:C:\Users\atypol\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar-56727 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
asasasasasaa
kiskiki
cherrycherry
aaaaa
Process finished with exit code 0
```

```
fun equation(expr: String): Int {
    val cleanExpr = expr.replace(" ", "")

    val tokens = mutableListOf<String>()
    val currentNumber = StringBuilder()

    for (char in cleanExpr) {
        if (char.isDigit()) {
            currentNumber.append(char)
        } else {
            if (currentNumber.isNotEmpty()) {
```

```

        tokens.add(currentNumber.toString())
        currentNumber.clear()
    }
    tokens.add(char.toString())
}

}

if (currentNumber.isNotEmpty()) {
    tokens.add(currentNumber.toString())
}

val processedTokens = mutableListOf<Any>()
var i = 0
while (i < tokens.size) {
    val token = tokens[i]
    when (token) {
        "*" -> {
            val prev = processedTokens.removeAt(processedTokens.size - 1)
as Int
            val next = tokens[i + 1].toInt()
            processedTokens.add(prev * next)
            i += 2
        }
        "/" -> {
as Int
            val prev = processedTokens.removeAt(processedTokens.size - 1)
            val next = tokens[i + 1].toInt()
            processedTokens.add(prev / next)
            i += 2
        }
        else -> {
            if (token.matches(Regex("\\d+"))) {
                processedTokens.add(token.toInt())
            } else {
                processedTokens.add(token)
            }
            i++
        }
    }
}

var result = processedTokens[0] as Int
i = 1
while (i < processedTokens.size) {
    val operator = processedTokens[i] as String
    val next = processedTokens[i + 1] as Int
    when (operator) {
        "+" -> result += next
        "-" -> result -= next
    }
    i += 2
}

return result
}

fun main() {
    println(equation("1+1"))
    println(equation("7*4-2"))
    println(equation("1+1+1+1+1"))
    println(equation("10/2+3"))
    println(equation("2+3*4"))
}

```

```
1 fun equation(expr: String): Int {
2     val tokens = expr.split(" ")
3     val processedTokens = mutableListOf<Int>()
4     var i = 0
5     while (i < tokens.size) {
6         if (tokens[i].matches(Regex("\\d+"))) {
7             processedTokens.add(tokens[i].toInt())
8         } else {
9             processedTokens.add(tokens[i])
10        }
11        i++
12    }
13    var result = processedTokens[0] as Int
14    while (i < processedTokens.size) {
15        val operator = processedTokens[i] as String
16        val next = processedTokens[i + 1] as Int
17        when (operator) {
18            "+" -> result += next
19            "-" -> result -= next
20        }
21        i += 2
22    }
23    return result
24}
```

Run

```
C:\Users\atypo\jdk\openjdk-23.0.1\bin\java.exe -javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56788 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
2
26
5
8
14
Process finished with exit code 0
```

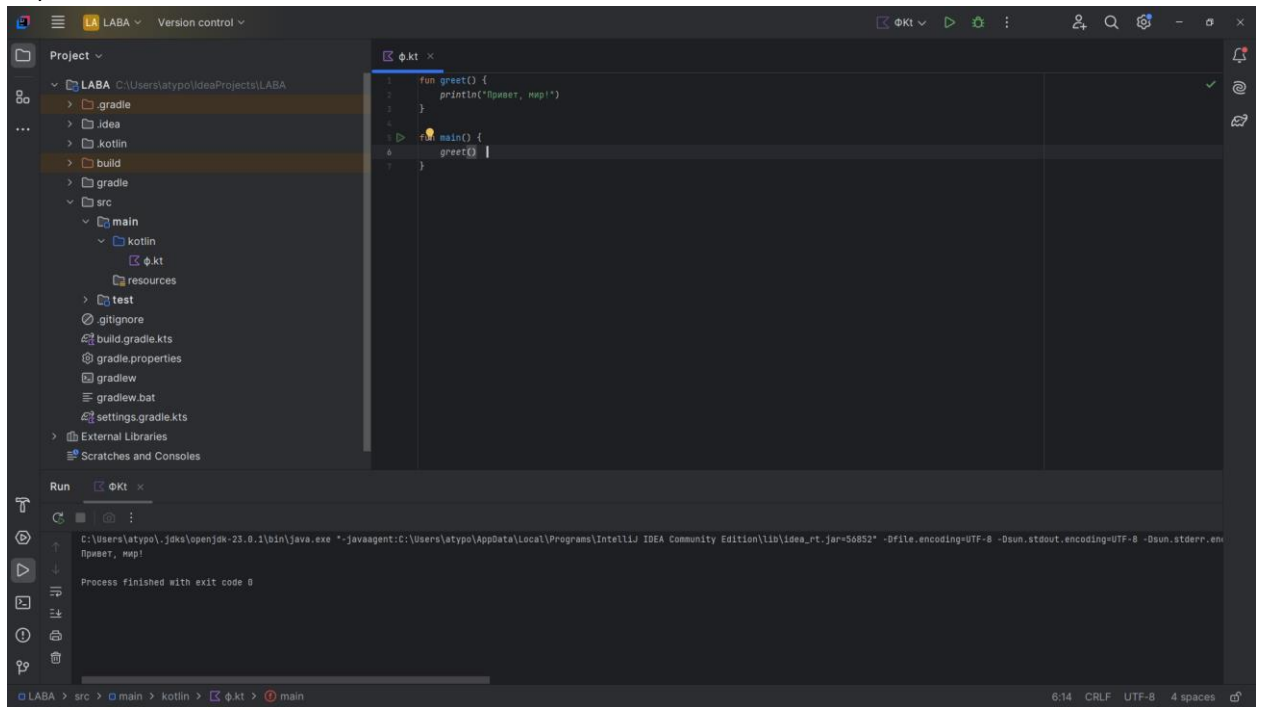
12)

```
1 fun google(number: Int): String {
2     return "o${"o".repeat(number)}gle"
3 }
4
5 fun main() {
6     println(google(5))
7     println(google(6))
8     println(google(2))
9 }
```

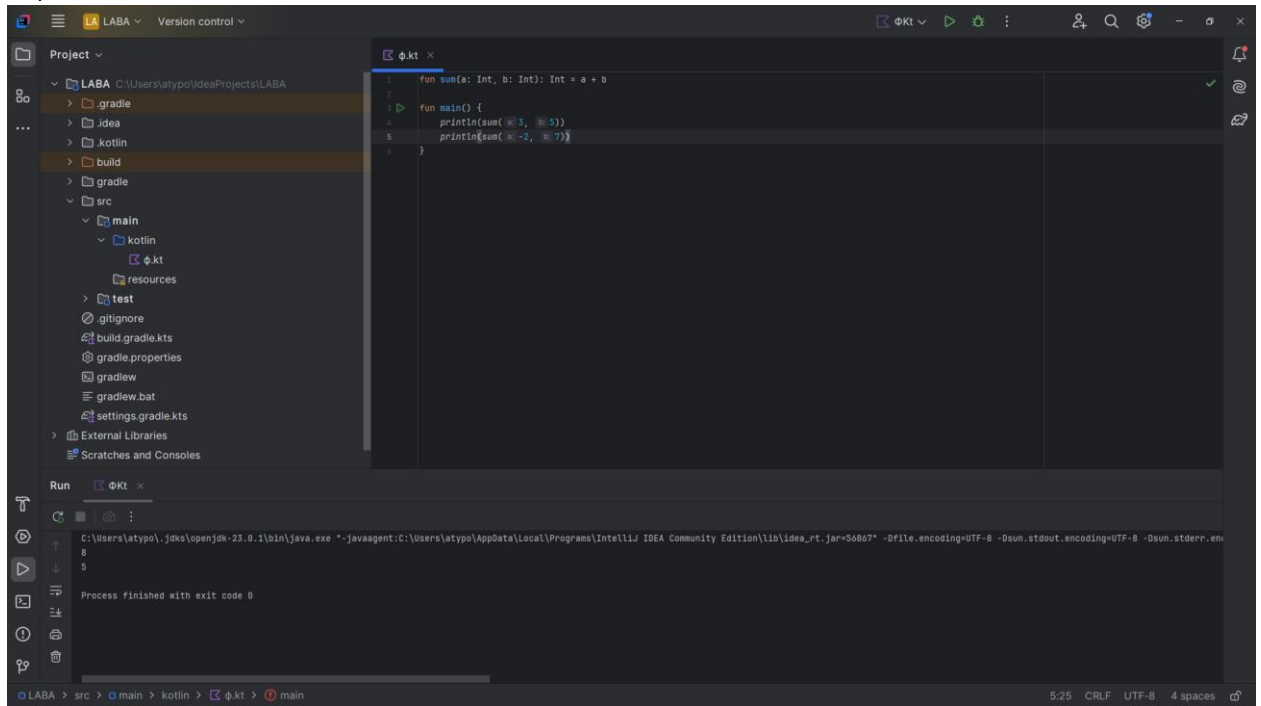
Run

```
C:\Users\atypo\jdk\openjdk-23.0.1\bin\java.exe -javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56822 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Google
Google
Google
Process finished with exit code 0
```

13)



14)



15)

The screenshot shows the IntelliJ IDEA interface with a Kotlin file open. The code defines a function `maxOfTwo` and a `main` function to test it.

```

1 fun maxOfTwo(a: Int, b: Int): Int = if (a > b) a else b
2
3 fun main() {
4     println(maxOfTwo(3, 5))
5     println(maxOfTwo(-2, -7))
6     println(maxOfTwo(10, 15))
7 }

```

The Run console shows the output of the program:

```

C:\Users\atypo\.jds\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56892" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.en
-2
10
Process finished with exit code 0

```

The status bar at the bottom indicates the file is at `LABA > src > main > kotlin > φ.kt > main` with a cursor at line 6, column 31.

16)

The screenshot shows the IntelliJ IDEA interface with a Kotlin file open. The code defines a function `isEven` and a `main` function to test it.

```

1 fun isEven(number: Int): Boolean = number % 2 == 0
2
3 fun main() {
4     println(isEven(4))
5     println(isEven(7))
6     println(isEven(0))
7     println(isEven(-3))
8     println(isEven(-8))
9 }

```

The Run console shows the output of the program:

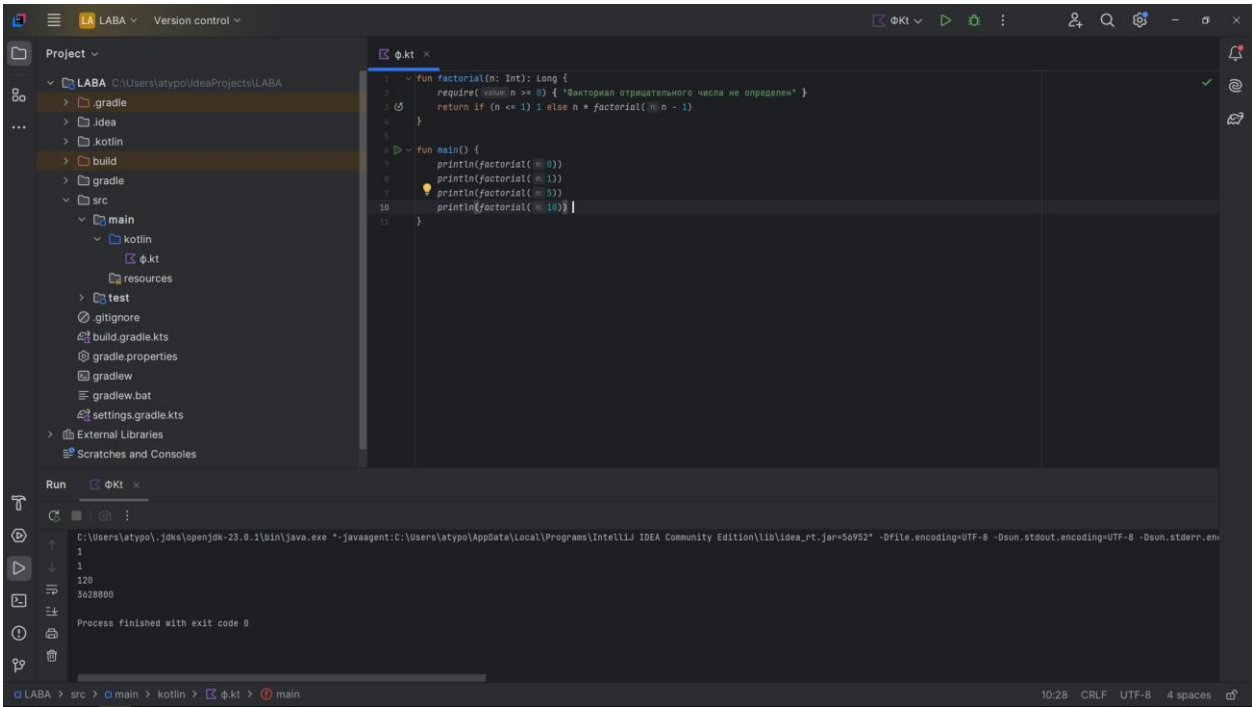
```

C:\Users\atypo\.jds\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\atypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=56920" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.en
true
false
true
false
true
Process finished with exit code 0

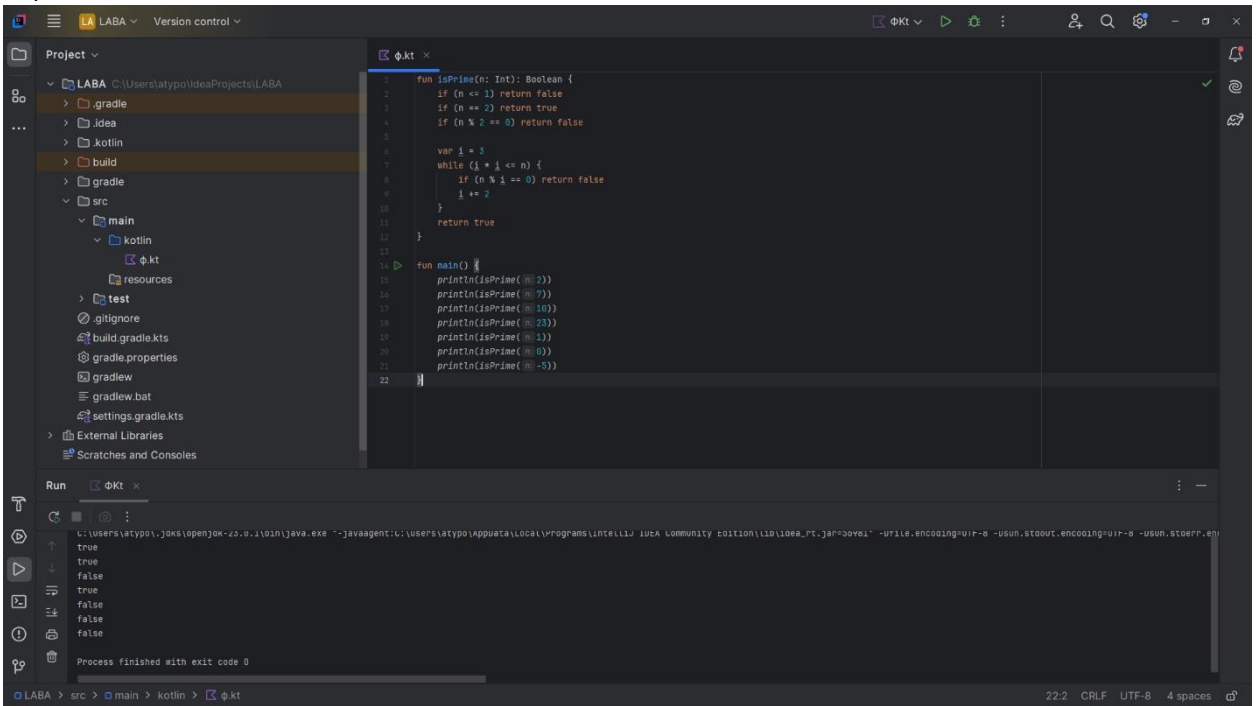
```

The status bar at the bottom indicates the file is at `LABA > src > main > kotlin > φ.kt > main` with a cursor at line 8, column 26.

17)



18)



19)

The screenshot shows the IntelliJ IDEA interface with a Kotlin file open. The code defines two functions: `sumArray` and `sumArrayManual`. `sumArray` is a simple lambda function that returns the sum of an array. `sumArrayManual` is a manual implementation that iterates through the array and calculates the sum. The `main` function tests both with arrays `arr1`, `arr2`, and `arr3`. The Run console shows the program executed successfully with exit code 0.

```

1 fun sumArray(numbers: IntArray): Int = numbers.sum()
2
3 fun sumArrayManual(numbers: IntArray): Int {
4     var sum = 0
5     for (num in numbers) {
6         sum += num
7     }
8     return sum
9 }
10
11 fun main() {
12     val arr1 = IntArrayOf(1, 2, 3, 4, 5)
13     val arr2 = IntArrayOf(-1, 0, 1)
14     val arr3 = IntArrayOf()
15
16     println(sumArray(arr1))
17     println(sumArrayManual(arr2))
18     println(sumArray(arr3))
19 }

```

Run console output:

```

C:\Users\ltypo\.jds\openjdk-23.0.1\bin\java.exe -javaagent:C:\Users\ltypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=57810* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.en
15
0
0
Process finished with exit code 0

```

20)

The screenshot shows the IntelliJ IDEA interface with a Kotlin file open. The code defines two functions: `findMax` and `findMaxManual`. `findMax` is a lambda function that returns the maximum value of an array, with a null check. `findMaxManual` is a manual implementation that iterates through the array and finds the maximum value. The `main` function tests both with arrays `arr1`, `arr2`, and `arr3`. The Run console shows the program executed successfully with exit code 0, but it also displays an exception message: "Массив не должен быть пустым" (Array must not be empty).

```

1 fun findMax(numbers: IntArray): Int = numbers.maxOrNull() ?: throw IllegalArgumentException("Массив не должен быть пустым")
2
3 fun findMaxManual(numbers: IntArray): Int {
4     require(numbers.isNotEmpty()) { "Массив не должен быть пустым" }
5     var max = numbers[0]
6     for (num in numbers) {
7         if (num > max) max = num
8     }
9     return max
10 }
11
12 fun main() {
13     val arr1 = IntArrayOf(1, 2, 3, 4, 5)
14     val arr2 = IntArrayOf(-1, -5, -3)
15     val arr3 = IntArrayOf()
16
17     println(findMax(arr1))
18     println(findMaxManual(arr2))
19     println(findMax(arr3))
20
21     try {
22         findMax(IntArrayOf())
23     } catch (e: IllegalArgumentException) {
24         println(e.message)
25     }
26 }

```

Run console output:

```

C:\Users\ltypo\.jds\openjdk-23.0.1\bin\java.exe -javaagent:C:\Users\ltypo\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=57857* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.en
9
-1
42
Массив не должен быть пустым
Process finished with exit code 0

```

21)

```

1 fun sortArray(numbers: IntArray): IntArray = numbers.sortedArray()
2
3 fun bubbleSort(numbers: IntArray): IntArray {
4     val sorted = numbers.copyOf()
5     for (i in sorted.indices) {
6         for (j in 0 until sorted.size - i - 1) {
7             if (sorted[j] > sorted[j + 1]) {
8                 val temp = sorted[j]
9                 sorted[j] = sorted[j + 1]
10                sorted[j + 1] = temp
11            }
12        }
13    }
14    return sorted
15 }
16
17 fun main() {
18     val arr1 = intArrayOf(1, 3, 0, 1, 2)
19     val arr2 = intArrayOf(-1, -3, 0, -5)
20     val arr3 = intArrayOf()
21
22     println(sortArray(arr1).contentToString())
23     println(bubbleSort(arr2).contentToString())
24     println(sortArray(arr3).contentToString())
25 }

```

Run console output:

```

C:\Users\laptop\jdk\openjdk-23.0.1\bin\java.exe "-javaagent:C:\Users\laptop\AppData\Local\Programs\IntelliJ IDEA Community Edition\lib\idea_rt.jar=57098" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
[1, 2, 3, 0, 1]
[-5, -3, -1, 0]
[]
Process finished with exit code 0

```

22)

```

fun isPalindrome(input: String): Boolean {
    val cleanInput = input.replace("[^a-zA-Za-яА-Я0-9]".toRegex(), "").lowercase()
    return cleanInput == cleanInput.reversed()
}

fun main() {
    val testString = "А роза упала на лапу Азора"
    println("'${testString}' является палиндромом? ${isPalindrome(testString)}")
}

```

Output:

```

'А роза упала на лапу Азора' является палиндромом? true

```

23)

The screenshot shows the Kotlin Playground interface. At the top, there's a header with the Kotlin logo and navigation links: Solutions, Docs, Community, Teach, Play, and a search icon. Below the header, there's a toolbar with version (2.1.20), JVM, and a dropdown menu showing 'Program arguments'. To the right of the dropdown are links for 'Copy link', 'Share code', and a 'Run' button. The main area contains the following Kotlin code:

```
fun countCharacters(input: String): Int {  
    return input.length  
}  
  
fun main() {  
    val testString = "Hello, мир!"  
    println("Количество символов в строке '$testString': ${countCharacters(testString)}")  
}
```

Below the code, the output is displayed: "Количество символов в строке 'Hello, мир!': 11". There are also icons for help (?) and close (X) on the right side of the output area.

24)

The screenshot shows the Kotlin Playground interface. At the top, there's a header with the Kotlin logo and navigation links: Solutions, Docs, Community, Teach, Play, and a search icon. Below the header, there's a toolbar with version (2.1.20), JVM, and a dropdown menu showing 'Program arguments'. To the right of the dropdown are links for 'Copy link', 'Share code', and a 'Run' button. The main area contains the following Kotlin code:

```
fun toUpperCase(input: String): String {  
    return input.uppercase()  
}  
  
fun main() {  
    val testString = "Привет, Kotlin!"  
    println("Исходная строка: '$testString'")  
    println("В верхнем регистре: '${toUpperCase(testString)}'")  
}
```

Below the code, the output is displayed: "Исходная строка: 'Привет, Kotlin!'" and "В верхнем регистре: 'ПРИВЕТ, KOTLIN!'". There are also icons for help (?) and close (X) on the right side of the output area.

25)

The screenshot shows the Kotlin Playground interface. At the top, there's a header with the Kotlin logo and navigation links: Solutions, Docs, Community, Teach, Play, and a search icon. Below the header, there's a toolbar with version (2.1.20), JVM, and a dropdown menu showing 'Program arguments'. To the right of the dropdown are links for 'Copy link', 'Share code', and a 'Run' button. The main area contains the following Kotlin code:

```
fun concatenateStrings(str1: String, str2: String): String {  
    return str1 + str2  
}  
  
fun main() {  
    val hello = "Привет, "  
    val world = "мир!"  
    println(concatenateStrings(hello, world))  
}
```

Below the code, the output is displayed: "Привет, мир!". There are also icons for help (?) and close (X) on the right side of the output area.

26)

Kotlin

SolutionsDocsCommunityTeachPlay

2.1.20JVMProgram argumentsCopy linkShare codeRun

```
fun <T> lastElement(array: Array<T>): T {
    return array.last()
}

fun main() {
    val numbers = arrayOf(1, 2, 3, 4, 5)
    val strings = arrayOf("Kotlin", "Java", "Python")
    val emptyArray = emptyArray<String>()

    println(lastElement(numbers))
    println(lastElement(strings))
}
```

5
Python

27)

Kotlin

SolutionsDocsCommunityTeachPlay

2.1.20JVMProgram argumentsCopy linkShare codeRun

```
fun <T>Array<T>.containsElement(element: T): Boolean {
    return this.contains(element)
}

fun main() {
    val colors = arrayOf("Red", "Green", "Blue")
    println(colors.containsElement("Green")) // true
    println(colors.containsElement("Yellow")) // false
}
```

true
false

28)

Kotlin

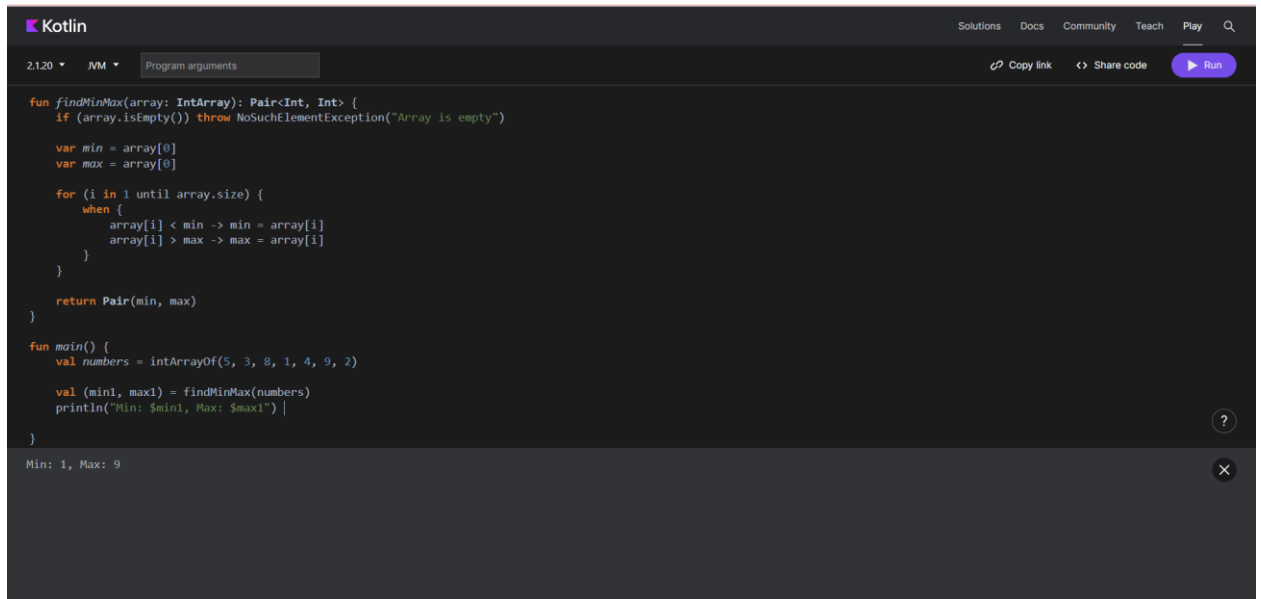
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2.1.20JVMProgram argumentsCopy linkShare codeRun

```
fun main() {
    val quickArray = (1..10).toList().toIntArray()
    println(quickArray.contentToString())
}
```

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

29)

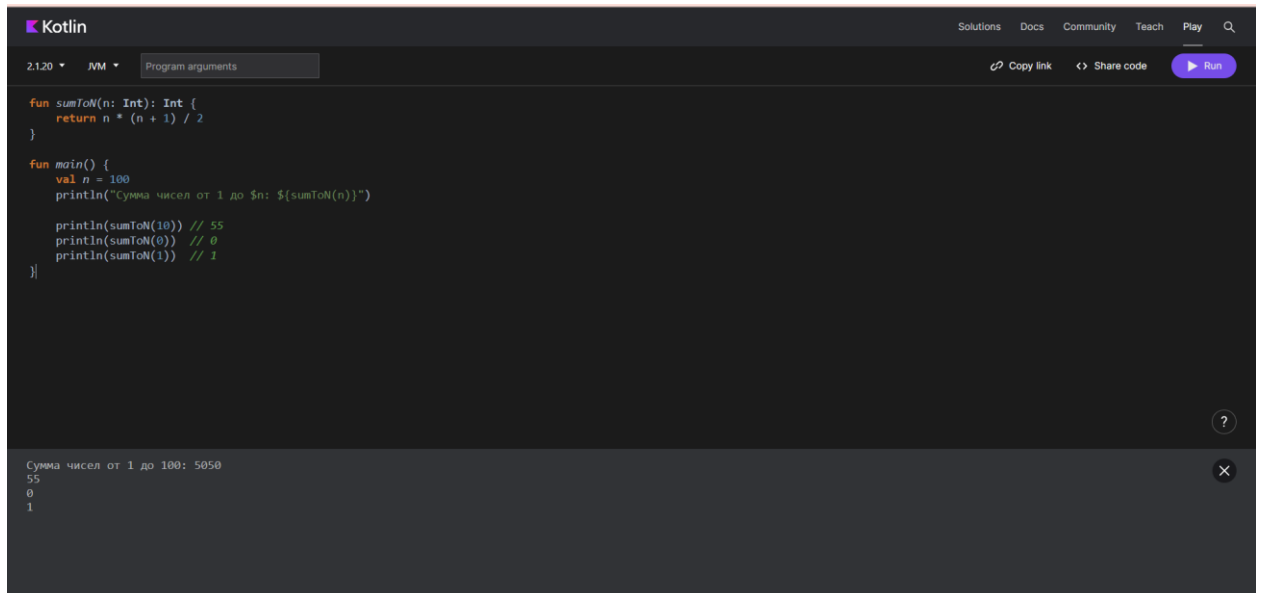


The screenshot shows the Kotlin IDE with a program to find the minimum and maximum values in an array. The code defines a function `findMinMax` that takes an `IntArray` and returns a `Pair<Int, Int>`. It checks for an empty array, initializes `min` and `max` to the first element, and then iterates through the array to update these values. The `main` function calls `findMinMax` on the array `[5, 3, 8, 1, 4, 9, 2]` and prints the results.

```
fun findMinMax(array: IntArray): Pair<Int, Int> {  
    if (array.isEmpty()) throw NoSuchElementException("Array is empty")  
  
    var min = array[0]  
    var max = array[0]  
  
    for (i in 1 until array.size) {  
        when {  
            array[i] < min -> min = array[i]  
            array[i] > max -> max = array[i]  
        }  
    }  
  
    return Pair(min, max)  
}  
  
fun main() {  
    val numbers = intArrayOf(5, 3, 8, 1, 4, 9, 2)  
  
    val (min1, max1) = findMinMax(numbers)  
    println("Min: $min1, Max: $max1")  
}
```

Min: 1, Max: 9

30)

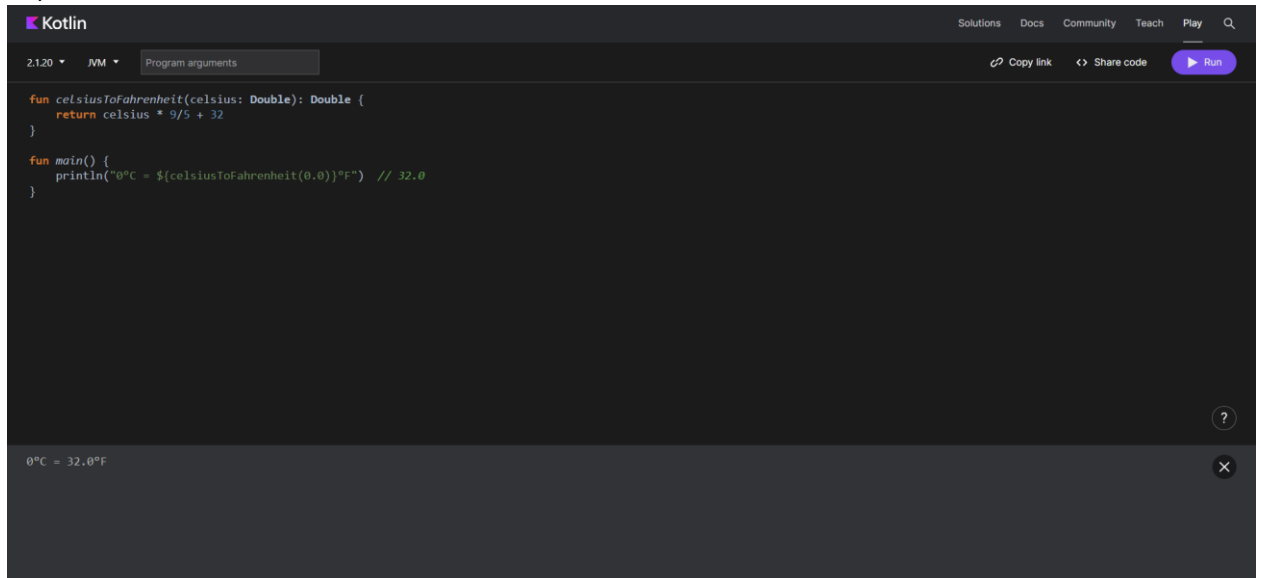


The screenshot shows the Kotlin IDE with a program to calculate the sum of numbers from 1 to `n`. The code defines a function `sumToN` that returns `n * (n + 1) / 2`. The `main` function sets `n = 100` and prints the sum. It also includes comments for the sum of numbers from 0 to 10, 0 to 0, and 1 to 1.

```
fun sumToN(n: Int): Int {  
    return n * (n + 1) / 2  
}  
  
fun main() {  
    val n = 100  
    println("Сумма чисел от 1 до $n: ${sumToN(n)}")  
  
    println(sumToN(10)) // 55  
    println(sumToN(0)) // 0  
    println(sumToN(1)) // 1  
}
```

Сумма чисел от 1 до 100: 5050
55
0
1

31)



The screenshot shows the Kotlin IDE with a program to convert Celsius to Fahrenheit. The code defines a function `celsiusToFahrenheit` that takes a `Double` and returns a `Double` calculated as `celsius * 9/5 + 32`. The `main` function prints the result of converting 0.0°C to Fahrenheit.

```
fun celsiusToFahrenheit(celsius: Double): Double {  
    return celsius * 9/5 + 32  
}  
  
fun main() {  
    println("0°C = ${celsiusToFahrenheit(0.0)}°F") // 32.0  
}
```

0°C = 32.0°F

32)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun reverseString(input: String): String {
    return input.reversed()
}
fun main() {
    val original = "Kotlin"

    println(reverseString(original))
    println(reverseString("Hello World"))
    println(reverseString("12345"))
}
```

niltoK
dlroW olleH
54321

33)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun <T> getElementAtIndex(array: Array<T>, index: Int): T {
    if (index < 0 || index >= array.size) {
        throw IndexOutOfBoundsException("Индекс $index выходит за границы массива (0..${array.size - 1})")
    }
    return array[index]
}
fun main() {
    val languages = arrayOf("Kotlin", "Java", "Python", "C++")
    val numbers = intArrayOf(10, 20, 30, 40, 50)

    println(getElementAtIndex(languages, 1)) // "Java"

    try {
        println(getElementAtIndex(languages, 10))
    } catch (e: IndexOutOfBoundsException) {
        println(e.message)
    }
}
```

Java
Индекс 10 выходит за границы массива (0..3)

34)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun removeSpaces(input: String): String {
    return input.replace(" ", "")
}
fun main() {
    val text = "K o t l i n   P r o g r a m m i n g"

    println(removeSpaces(text))

    val sentence = "Remove all spaces from this sentence"
    println(removeSpaces(sentence))
}
```

KotlinProgramming
Removeallspacesfromthissentence

35)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun sumFirstNNumbers(n: Int): Int {
    return n * (n + 1) / 2
}

fun main() {
    println(sumFirstNNumbers(5)) // 15 (1+2+3+4+5)
    println(sumFirstNNumbers(10)) // 55
    println(sumFirstNNumbers(0)) // 0
    println(sumFirstNNumbers(1)) // 1
}
```

15
55
0
1

36)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun String.containsSubString(subString: String, ignoreCase: Boolean = false): Boolean {
    return this.contains(subString, ignoreCase)
}

fun main() {
    val text = "Kotlin - современный язык программирования"
    println(text.containsSubString("язык")) // true
}
```

true

37)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun printMultiplicationTable(number: Int) {  
    println("Таблица умножения для числа $number:")  
    for (i in 1..10) {  
        println("$number x $i = ${number * i}")  
    }  
}  
fun main() {  
    printMultiplicationTable(5)  
}
```

?

Таблица умножения для числа 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

X

38)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun getStringLength(input: String): Int {  
    return input.length  
}  
fun main() {  
    val text = "Hello, Kotlin!"  
    val nullableText: String? = null  
    println("Длина строки: ${getStringLength(text)}")  
}
```

?

Длина строки: 14

X

39)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun reverseArray(arr: IntArray): IntArray {  
    return arr.reversedArray()  
}  
fun main() {  
    val original = intArrayOf(1, 2, 3, 4, 5)  
    val reversed1 = reverseArray(original)  
    println(reversed1.contentToString())  
}
```

?

[5, 4, 3, 2, 1]

X

40)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun copyIntArray(arr: IntArray): IntArray {
    return arr.copyOf()
}

fun main() {
    val original = intArrayOf(1, 2, 3, 4, 5)
    println(original.contentToString())

    val copy1 = copyIntArray(original)
    println(copy1.contentToString())
}
```

[1, 2, 3, 4, 5]
[1, 2, 3, 4, 5]

41)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun countVowels(input: String): Int {
    val vowels = setOf('a', 'e', 'i', 'o', 'u', 'a', 'y', 'o', 'u', 'w', 'a', 'n', 'w', 'e', 'e')
    return input.lowercase().count { it in vowels }
}

fun main() {
    val text = "Привет, Kotlin! Hello World!"
    println("Количество гласных: ${countVowels(text)}")
}
```

Количество гласных: 7

42)

Kotlin

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2.1.20 JVM Program arguments

Copy link Share code Run

```
fun findFirstIndex(arr: IntArray, target: Int): Int {
    return arr.indexOf(target)
}

fun main() {
    val numbers = intArrayOf(3, 7, 2, 9, 5, 7, 1)
    val names = arrayOf("Alice", "Bob", "Charlie", "Alice")

    println(findFirstIndex(numbers, 7)) // 1
    println(findFirstIndex(numbers, 4)) // -1
}
```

1
-1